Chapter 12 Answers

Problem 1

(a): Take N to be Vft, as $Vftt = tft = f$, and $Vftf = fft = t$.

(b): Take $\wedge$ to be Ff, as $Fftt = ttf = t$, $Fftf = ftf = f$, $Ffft = tff = f$, and $Ffff = fff = f$. Smullyan gives an alternative solution, taking $\wedge$ to be Rf, as $Rfpq = pqf$ (which can easily be seen to also work).

(c): Take $\vee$ to be Vt, as $Vttt = ttt = t$, $Vttf = ftt = t$, $Vtft = ttf = t$, and $Vtff = ftf = f$. Smullyan gives an alternative solution, taking $\vee$ to be Tt, which can easily be seen to work.

(d): Take $\supset$ to Rt, as $Rttt = ttt = t$, $Rttf = tft = f$, $Rtft = ftt = t$, and $Rtff = fft = t$.

(e): Take $\equiv$ to be the combinator

$$W(W_2(F^{**}(R^*(R^*(B_1E\ \supset)\wedge)\ \supset)))$$

as

$$\begin{aligned}
&W(W_2(F^{**}(R^*(R^*(B_1E\ \supset)\wedge)\ \supset)))pq\\
&= W_2(F^{**}(R^*(R^*(B_1E\ \supset)\wedge)\ \supset))ppq\\
&= F^{**}(R^*(R^*(B_1E\ \supset)\wedge)\ \supset)ppqq\\
&= R^*(R^*(B_1E\ \supset)\wedge)\ \supset pqqp\\
&= R^*(B_1E\ \supset)\wedge pq\ \supset qp\\
&= B_1E\ \supset pq\wedge\ \supset qp\\
&= E(\supset pq)\wedge\ \supset qp\\
&= (\supset pq)\wedge(\supset qp)
\end{aligned}$$

That is, $\equiv pq = (\supset pq) \wedge (\supset qp)$, which is how we wished to define $\equiv$.

Smullyan provides a far simpler solution for part (e). He notes that we want $\equiv$ to be such that:

$$\equiv pq = pq(Nq)$$

which can be received by taking $\equiv$ to be the combinator CSN, as

$$CSNpq = SpNq = pq(Nq)$$

Problem 2

Let n^+ be an abbreviation for $n+1$. We first show that, for every n, $\overline{0} \neq \overline{n^+}$. Suppose that $\overline{0} = \overline{n^+}$. Then $I = Vf\sigma$, meaning $K = IK = V(KI)\sigma K = K(KI)\sigma = KI$, which is in contradiction to Problem 19 of Chapter 8, which proved that $Kx \neq K$ for all x. Next, we show that, if $\overline{m^+} = \overline{n^+}$, then $\overline{m} = \overline{n}$. If $\overline{m^+} = \overline{n^+}$, then $Vf\overline{m} = Vf\overline{n}$, meaning $\overline{m} = ff\overline{m} = Vf\overline{m}f = Vf\overline{n}f = ff\overline{n} = \overline{n}$. Lastly, we show that, for all n and positive k, $\overline{n} \neq \overline{n+k}$. But suppose $\overline{n} = \overline{n+k}$. Now, by the first result of this problem, we know that $n > 0$. Consider the sequence of combinators

$$\overline{n-1}, \overline{n-2}, ..., \overline{n-(n-1)}, \overline{n-n}$$

For each $\overline{n-i}$, it is true that $\overline{n-i} = \overline{(n+k)-i}$. For, because $\overline{n} = \overline{n+k}$, then $\overline{n-1} = \overline{(n+k)-1}$ by the second result of this problem. Then, if $\overline{n-i} = \overline{(n+k)-i}$ holds for some i, and there is an $i+1$th term in the sequence, $\overline{n-(i+1)} = \overline{(n+k)-(i+1)}$ again by the second result of the problem. Thus $\overline{0} = \overline{n-n} = \overline{(n+k)-n}$ such that $(n+k)-n$ is positive, which is in contradiction to the first result of this problem. Therefore $\overline{n} \neq \overline{n+k}$. Thus, if $n \neq m$, either $m < n$ or $n < m$. If the former, then the preceeding result would show that $\overline{m} \neq \overline{n}$, which is the same as saying $\overline{n} \neq \overline{m}$. Similarly, if $n < m$, then the preceeding result proves that $\overline{n} \neq \overline{m}$, which is the same as saying $\overline{m} \neq \overline{n}$.

Problem 3

Take P to be $T(CK)$, as

$$T(CK)\overline{n^+} = T(CK)(Vf\bar{n}) = Vf\bar{n}(CK) = CKf\bar{n} = K\bar{n}f = \bar{n}$$

Smullyan provides an alternative solution. He takes P to be Tf, as $Tf\overline{n^+} = Vf\bar{n}f = ff\bar{n} = \bar{n}$.

Problem 4

Take Z to be TK (that is, Tt), as $TK\bar{0} = TKI = IK = K = t$, and $TK\overline{n^+} = TK(Vf\bar{n}) = Vf\bar{n}K = Kf\bar{n} = f$ for all n.

Problem 5

This is just the combinator Z from Problem 4, as $Z\bar{0}xy = txy = x$ and $Z\overline{n^+}xy = fxy = y$.

Problem 6

Consider any combinator A and number k. Consider the realizable term $t(x, x_1) = Zx_1\bar{k}(A(x(Px_1)))$. By the fixed point theorem, there is a combinator Γ such that $\Gamma x_1 = t(\Gamma, x_1) = Zx_1\bar{k}(A(\Gamma(Px_1)))$. Thus

$$\Gamma\bar{0} = Z\bar{0}\bar{k}(A(\Gamma(P\bar{0}))) = t\bar{k}(A(\Gamma(P\bar{0}))) = \bar{k}$$

and

$$\Gamma\overline{n^+} = Z\overline{n^+}\bar{k}(A(\Gamma(P\overline{n^+})))$$

$$= f\bar{k}(A(\Gamma\bar{n}))$$
$$= A(\Gamma\bar{n})$$

Note that if θ is the combinator that realizes the term $t(x, x_1)$, then Γ is specifically the fixed point of θ.

Problem 7

(a): We can use a similar concept as in Problem 6. Now, however, we want a combinator Γ such that for any combinator A:
C_1: $\Gamma\bar{0}\bar{k} = \bar{k}$.
C_2: $\Gamma\overline{n^+}\bar{k} = A(\Gamma\bar{n}\bar{k})$.
We can find this combinator by, in Problem 6, instead letting $t(x, x_1, x_2)$ be the term $Zx_1x_2(A(x(Px_1)x_2))$, and then taking Γ to be the Γ from the fixed point theorem such that $\Gamma x_1x_2 = t(\Gamma, x_1, x_2)$. Now, let A be the combinator $V(KI)$. We then take $\bigoplus$ to be the Γ such that $\Gamma x_1x_2 = Zx_1x_2(V(KI)(\Gamma(Px_1)x_2))$ (Smullyan's answer is slightly different: he takes $\Gamma x_1x_2 = Zx_2x_1(V(KI)(\Gamma x_1(Px_2)))$, which could be proven by doing induction on n instead of m), meaning $\Gamma\bar{0}\bar{k} = \bar{k}$ and $\Gamma\bar{m}^+\bar{k} = V(KI)(\Gamma\bar{m}\bar{k})$. Then, we can show that $\Gamma\overline{m}n = \overline{m^+ + n}$ by induction. Now, the base case when $m = 0$ holds, as $\Gamma\bar{0}\bar{n} = \bar{n} = \overline{0 + n}$. Then suppose the hypothesis holds for some m, that is, $\Gamma\overline{mn} = \overline{m + n}$. Then

$$\Gamma\overline{m^+}n = V(KI)(\Gamma\bar{m}\bar{n}) = V(KI)\overline{m + n} = \overline{(m + n) + 1}$$

Thus, in all cases, $\Gamma\bar{m}\bar{n} = \overline{m + n}$.

(b): This can be solved again by making a variation on Problem 6. Consider the realizable term $t(x, x_1, x_2) = Zx_2\bar{0}(\bigoplus x_1(xx_1(Px_2)))$ (It can easily be seen by similar reasoning as below that we could also take $t(x, x_1, x_2) = Zx_1\bar{0}(\bigoplus x_1(xx_1(Px_2)))$ to be the realizable term, only then doing induction on m, not n). By the fixed point theorem, there is some Γ such that $\Gamma x_1x_2 = Zx_2\bar{0}(\bigoplus x_1(\Gamma x_1(Px_2)))$. Note that

$$\Gamma\bar{m}\bar{0} = Z\bar{0}\bar{0}(\bigoplus \bar{m}(\Gamma\bar{m}(P\bar{0}))) = \bar{0}$$

and

$$\Gamma\bar{m}\overline{n^+} = Z\overline{n^+}\bar{0}(\bigoplus \bar{m}(\Gamma\bar{m}(P\overline{n^+})))$$
$$= \bigoplus \bar{m}(\Gamma\bar{m}\bar{n})$$

This is the combinator we take for $\bigotimes$. Now, we can show that this combinator works by induction on n. The base case is when $n = 0$. Then, as we have seen, $\Gamma\bar{m}\bar{n} = \Gamma\bar{m}\bar{0} = \bar{0} = \overline{n \times 0}$. Now suppose the hypothesis holds for some n, that is, $\Gamma\bar{m}\bar{n} = \overline{m \times n}$. Then

$$\Gamma\bar{m}\overline{n^+} = \bigoplus \bar{m}(\Gamma\bar{m}\bar{n}) = \bigoplus \bar{m}\overline{m \times n}$$
$$= \overline{m + (m \times n)} = \overline{m \times (n + 1)}$$

(c):Ⓔ Note that this answer assumes $m > 0$. Additionaly note that Smullyans answer seems incorrect. His yields Ⓔ$\bar{0}\bar{m} = 1$, which should not happen (as, for positive m, $0^m = 0$). We can yet again make a variation of Problem 6. Consider the term $t(x, x_1, x_2) = Zx_2\bar{1}(\bigotimes x_1(xx_1(Px_2)))$. By the fixed point theorem, there is some Γ such that $\Gamma x_1x_2 = Zx_2\bar{1}(\bigoplus x_1(\Gamma x_1(Px_2)))$. Note that $\Gamma\bar{m}\bar{0} = \bar{1}$, and $\Gamma\bar{m}\overline{n^+} = \bigotimes x_1(\Gamma\bar{m}\bar{n})$. This is the combinator we take for Ⓔ Now we can prove by induction that $\Gamma\bar{m}\bar{n} = \overline{m^n}$. The base case, when $n = 0$, was noted above. Now suppose the hypothesis holds for some n. Then

$$\Gamma\bar{m}\overline{n^+} = \bigotimes \bar{m}(\Gamma\bar{m}\bar{n}) = \bigotimes \bar{m}\overline{m^n}$$
$$= \overline{m \times m^n} = \overline{m^{n^+}}$$

Problem 8

Consider the realizable term $t(x, x_1) = Zx_1t(x(Px_1)ft)$. By the fixed point theorem, there is a Γ such that $\Gamma\bar{n} = Z\bar{n}t(\Gamma(P\bar{n})ft)$. This obviously works; if $n = 0$, $\Gamma\bar{0} = Z\bar{0}t(\Gamma(P\bar{n})ft) = t$, and if $n > 0$, $\Gamma\bar{n} = Z\bar{n}t(\Gamma(P\bar{n})ft) = \Gamma(P\bar{n})ft$, that is, if $n-1$ is even, $\Gamma\bar{n} = f$, and if $n-1$ is odd, $\Gamma\bar{n} = t$, which is the condition we desired.

Problem 9

In fact, it is not $N\Gamma$ that computes $\bar{A}$, but $BN\Gamma$. For suppose $n \in A$. Then $\Gamma\bar{n} = t$, meaning $BN\Gamma\bar{n} = N(\Gamma\bar{n}) = Nt = f$. If $n \notin A$, then $\Gamma\bar{n} = f$, meaning $BN\Gamma\bar{n} = N(\Gamma\bar{n}) = Nf = t$.

Problem 10

Note the importance of specifying that a relational combinator of degree n must always be such that $A\overline{k_1}...\overline{k_n} = t$ or $A\overline{k_1}...\overline{k_n} = f$. If this wasn't specified, it could be that it neither equaled t nor f. Now suppose we had such a relational combinator computing some relation R. Then if $R(k_1, ..., k_n)$ holds, $A\overline{k_1}...\overline{k_n} = t$, and if it doesn't hold, then $A\overline{k_1}...\overline{k_n} = f$. But the contrapositives of those two statements show that, if $A\overline{k_1}...\overline{k_2}$ does not equal either t or f, $R(k_1, ..., k_n)$ both holds and does not hold at the same time, a contradiction. Thus the requirenment that $A\overline{k_1}...\overline{k_n}$ is always either equal to t or f is a necessary one.

Consider the realizable term:

$$t(x, x_1x_2) = \wedge(N(Z\overline{x_1}))(\vee(Z\overline{x_2})(x(P\overline{x_1})(P\overline{x_2})))$$

By the fixed point theorem, there is some Γ such that:

$$\Gamma\bar{x}\bar{y} = \wedge(N(Z\bar{x}))(\vee(Z\bar{y})(\Gamma(P\bar{x})(P\bar{y})))$$

We can prove that this works by induction on x. The base case is when $x = 0$. Thus, because $N(Z\bar{0}) = f$,

$$\Gamma\bar{0}\bar{y} = \wedge(N(Z\bar{0}))(\vee(Z\bar{y})(\Gamma(P\bar{0})(P\bar{y}))) = f$$

Now suppose the hypothesis holds for some x, that is, for all y, if $x > y$, then $\Gamma xy = t$, and if $x \leq y$, then $\Gamma xy = f$. On the one hand, suppose that $x > y$. Then either $y = 0$, meaning $Z\bar{y} = t$, or $x - 1 > y - 1$, in which case $\Gamma(P\bar{x})(P\bar{y}) = t$ by the assumption. Thus, either of $Z\bar{y} = t$ or $\Gamma(P\bar{x})(P\bar{y}) = t$, and $N(Z\bar{x}) = t$ as well, as $x > 0$. Thus

$$\vee(Z\bar{y})(\Gamma(P\bar{x})(P\bar{y})) = t \text{ and } N(Z\bar{x}) = t$$

meaning

$$\Gamma\bar{x}\bar{y} = \wedge(N(Z\bar{x}))(\vee(Z\bar{y})(\Gamma(P\bar{x})(P\bar{y}))) = t$$

On the other hand, suppose that $x \leq y$. Thus $y \neq 0$ (and we are supposing that $x > 0$, as the case where $x = 0$ was proven above) and $x - 1 \leq y - 1$. Thus, by the assumption,

$$\vee(Z\bar{y})(\Gamma(P\bar{x})(P\bar{y})) = f$$

meaning

$$\Gamma\bar{x}\bar{y} = \wedge(N(Z\bar{x}))(\vee(Z\bar{y})(\Gamma(P\bar{x})(P\bar{y}))) = f$$

Smullyan provides an alternative solution which does not involve the logical combinators. By the fixed point theorem, there is some Γ such that $\Gamma xy = Zxf(Zyt(\Gamma(Px)(Py)))$. Γ can be seen to satisfy the conditions of the problem.

Problem 11

Suppose $f(x)$ and $g(x)$ are realizable by the combinators A_1 and A_2, respectively. Take the combinator A_3 which realizes $f(g(x))$ to be BA_1A_2, as $BA_1A_2\overline{n} = A_1(A_2\overline{n}) = A_1\overline{g(n)} = \overline{f(g(n))}$. In a similar fashion, we could take the combinator BA_2A_1 to be the combinator which realizes $g(f(x))$, as $BA_2A_1\overline{n} = A_2(A_1\overline{n}) = A_2\overline{f(n)} = \overline{g(f(n))}$.

Problem 12

We suppose that $R(x, y)$ is computable and that $f(x)$ and $g(x)$ are both realizable. Thus there is some combinator A such that if $R(x, y)$, then $A\bar{x}\bar{y} = t$, and if $R(x, y)$ doesn't hold, then $A\bar{x}\bar{y} = f$. Also, there are combinators A_1 and A_2 which realize $f(x)$ and $g(x)$, respectively. Then:

(a): BAA_1 computes $R(f(x), y)$, as if $R(f(x), y)$, then $BAA_1\bar{x}\bar{y} = A(A_1\bar{x})\bar{y} = A\overline{f(x)}\bar{y} = t$. If $R(f(x), y)$ does not hold, then $BAA_1\bar{x}\bar{y} = A(A_1\bar{x}) = A\overline{f(x)}\bar{y} = f$.

(b): Let A_3 be the combinator such that $A_3xyz = yxz$. Similar reasoning as in part (a) shows that $A_3^{**}DAA_2$ computes $R(x, g(y))$ (as $A_3^{**}DAA_2\bar{x}\bar{y} = DA\bar{x}A_2\bar{y} = A\bar{x}(A_2\bar{y})$). Smullyan provides an alternative solution: Take $BCDAA_2$ to be the combinator which computes the relation $R(x, g(y))$.

(c): Similar reasoning as in part (a) shows that $B(A_3^*(DA)A_2)A_1$ computes $R(f(x), g(y))$, as

$$B(A_3^*(DA)A_2)A_1\bar{x}\bar{y} = A_3^*(DA)A_2(A_1\bar{x})\bar{y}$$

$$= DA(A_1\bar{x})A_2\bar{y} = A(A_1\bar{x})(A_2\bar{y})$$

For part (c), Smullyan provides two swifter proofs. Let $S(x, y)$ be the relation $R(f(x), y)$. By part (a), BAA_1 computes it. Hence, by part (b), $S(x, g(y))$, that is, $R(f(x), g(y))$, is computed by $BCD(BAA_1)A_2$. Alternatively, let $\delta(x, y)$ be the relation $R(x, g(y))$. By part (b), $BCDAA_2$ computes it. Hence, by part (a), $\delta(f(x), y)$, that is, $R(f(x), g(y))$, is computed by $B(BCDAA_2)A_1$.

Problem 13

Suppose we have a regular relational combinator A of degree 2 (in fact, the result of this problem does not rely upon A being regular). Now consider the realizable term $t(x, x_1, x_2) = Ax_1x_2x_2(xx_1(\sigma y_1))$. By the fixed point theorem, we have some Γ such that $\Gamma\bar{x}\bar{y} = A\bar{x}\bar{y}\bar{y}(\Gamma\bar{x}(\sigma\bar{y}))$. Now suppose that $A\bar{n}\bar{m} = t$. Then

$$\Gamma\bar{n}\bar{m} = A\bar{n}\bar{m}\bar{m}(\Gamma\bar{n}(\sigma\bar{m})) = t\bar{m}(\Gamma\bar{n}(\sigma\bar{m})) = \bar{m}$$

On the other hand, suppose that $A\bar{n}\bar{m} = f$. Then

$$\Gamma\bar{n}\bar{m} = A\bar{n}\bar{m}\bar{m}(\Gamma\bar{n}(\sigma\bar{m})) = f\bar{m}(\Gamma\bar{n}(\sigma\bar{m}))$$

$$= \Gamma\overline{\bar{n}m + 1}$$

thus proving Γ satisfies the desired conditions of the problem.

Problem 14

Consider some regular relational combinator A of degree 2. Now let A' be the combinator $CA_1\bar{0}$, as $CA_1\bar{0}\bar{n} = A_1\bar{n}\bar{0}$. This obviously works, as $A_1\bar{n}\bar{0} = A_1\bar{n}\bar{1} = ... = A_1\bar{n}\bar{k} = \bar{k}$ for k the smallest number for which $A\bar{n}\bar{k} = t$.

Problem 15

By the fixed point theorem, there is some Γ such that $\Gamma xy = ((\textcircled{E}\overline{10}x) > y)x(\Gamma(\sigma x)y)$, that is,

$$\Gamma\bar{m}\bar{n} = ((\textcircled{E}\overline{10}\bar{m}) > \bar{n})\bar{m}(\Gamma\overline{m^+}\bar{n})$$

Then take L' to be $\Gamma\bar{1}$, which is a length measurer (as we recall that the length of n is the smallest number k such that $10^k > n$). Note that $(\textcircled{E}\overline{10}\bar{m}) > \bar{n}$ is an abbreviation for $g(\textcircled{E}\overline{10}\bar{m})\bar{n}$, which was defined in Problem 10.

Problem 16

Note that $x * y = (x \times 10^k) + y$ where k is the length of y. Thus take $\circledast$ to be the combinator which realizes the term

$$t(x, y) = \bigoplus(\bigotimes x(\textcircled{E}\overline{10}(L'y)))y$$

Problem 17

Let s be the Gödel number of Vf. We note that $0^{\#} = 3312424$ and $1^{\#}$ is the Gödel number of $(\sigma\bar{0})$, that is, $3 * s * 0^{\#} * 4$, and in general, $(n+1)^{\#}$ is the Gödel number of $(\sigma\bar{n})$, which is $3 * s * n^{\#} * 4$. Thus we take δ to be the combinator satisfying (recalling that Z is the zero tester and P is the predecessor combinator):

$$\delta\bar{n} = Z\bar{n}\overline{3312424}(\bar{3} * \bar{s} * (\delta(P\bar{n})) * \bar{4})$$

which exists by the fixed point theorem. Note that $(\bar{3} * \bar{s} * (\delta(P\bar{n})) * \bar{4})$ is an abbreviation for

$$\circledast\bar{3}\,(\circledast\bar{s}\,(\circledast(\delta(P\bar{n}))\bar{4}))$$

An interesting alternative derivation of δ can be found in the following way: Let A be the combinator satisfying the condition: $A\bar{n} = \bar{3} * \bar{s} * \bar{n} * \bar{4}$. By Problem 6, there is a combinator Γ (in fact, the Γ obtained in this way is the same as the one in the preceeding paragraph) satisfying the following conditions:
C_1: $\Gamma\bar{0} = \overline{3312424}$
C_2: $\Gamma\overline{n^+} = A(\Gamma\bar{n})$
We prove by induction that we can take δ to be Γ. The base case is given by C_1. Now suppose it works for some n, that is, $\Gamma\overline{n} = \overline{n^{\#}}$. Then

$$\Gamma\overline{n^+} = A(\Gamma\bar{n}) = A\overline{n^{\#}} = \bar{3} * \bar{s} * \overline{n^{\#}} * \bar{4} = \overline{(n^+)^{\#}}$$

thus completing the proof.

Problem 18

This is just the combinator Δ satisfying the condition:

$$\Delta\bar{n} = \circledast\bar{n}(\delta\bar{n})$$

We could take Δ to be $S\circledast\delta$ (as can be easily verified). Alternatively, we could take Δ to be $W(DC\circledast\delta)$. This is because $Sxyz = W(DCxy)z$. For $Sxyz = xz(yz)$ and

$$W(DCxy)z = DCxyzz = Cx(yz)z = xz(yz)$$

Problem 19

(b) is the statement which is true (for all numerical combinators X). For $\Delta\lceil X\rceil = \overline{n * n^{\#}}$ where n is the Gödel number of X. Now, $X\lceil X\rceil = n{*}n^{\#}$, that is, the norm of X. Then, by definition, $\lceil X\lceil X\rceil\rceil = \overline{n * n^{\#}}$, thus showing $\Delta\lceil X\rceil = \lceil X\lceil X\rceil\rceil$.

Problem 20

Consider any term A. Let X be the term $A(\Delta\lceil BA\Delta\rceil)$, as

$$\begin{aligned} X = A(\Delta\lceil BA\Delta\rceil) &= A\lceil BA\Delta\lceil BA\Delta\rceil\rceil \\ &= A\lceil A(\Delta\lceil BA\Delta\rceil)\rceil \end{aligned}$$

and

$$A\lceil X\rceil = A\lceil A(\Delta\lceil BA\Delta\rceil)\rceil$$

that is, $A\lceil X\rceil = X$.

Problem 21

This is immediate from the second fixed point theorem. For suppose we have some realizable function $f(x)$. Hence it is realized by some combinator A. By the second fixed point theorem, there is some term X such that $A\lceil X\rceil = X$. But we also know that $A\bar{n} = \overline{f(n)}$ for all n. Thus, because $\lceil X\rceil$ is some $\bar{n}$ for n the Gödel number of X, $A\lceil X\rceil = \overline{f(n)}$. Thus $X = \overline{f(n)}$, that is, X designates $f(n)$, where n is the Gödel number of X.

Problem 22

(a): This is obviously true. For suppose Γ computes A. Thus $\Gamma\bar{n} = t$ for every n in A, and $\Gamma\bar{n} = f$ for every n that is not in A. To show that Γ represents A, we need to show that $\Gamma\bar{n} = t$ iff $n \in A$. If $n \in A$, we are already given that $\Gamma\bar{n} = t$. Now suppose that $\Gamma\bar{n} = t$. Then if $n \notin A$, we would have $\Gamma\bar{n} = f$. But $t \neq f$, as this would contradict Problem 19 of Chapter 8, which proved that $K \neq Kx$ for every x (we recall that $t = K$ and $f = KI$).

(b): By Problem 9, $BN\Gamma$ computes $\bar{A}$. Thus, by part (a) of this problem, $\bar{A}$ is also representable.

Problem 23

Suppose $f(x)$ is realizable by Γ, and A is representable by the combinator Y. Thus $\Gamma\bar{n} = \overline{f(n)}$ and if $n \in A$, then $Y\bar{n} = t$, and if $n \notin A$, then $Y\bar{n} = f$. Let X be the combinator realizing the term $t(x) = Y(\Gamma x)$, that is, $X\bar{n} = Y(\Gamma\bar{n})$. X represents $f^{-1}(A)$. For suppose $n \in f^{-1}(A)$. Then $f(n) \in A$. Hence

$$X\bar{n} = Y(\Gamma\bar{n}) = Y\overline{f(n)} = t$$

On the other hand, suppose $n \notin f^{-1}(A)$. Then $f(n) \notin A$. Hence

$$X\bar{n} = Y(\Gamma\bar{n}) = Y\overline{f(n)} = f$$

Problem 24

Suppose that A is representable by Γ. Now let f be the function such that $f(x) = x * 52$. This is realizable by the combinator γ realizing the term $t(x) = \circledast\bar{x}\overline{52}$. Then, by Problem 23, $f^{-1}(A)$ is representable. That is, the set of all n such that $(n * 52) \in A$ is representable.

Problem 25

For this problem, note that if n is the Gödel number of X, then $n*52$ is the Gödel number of $X = t$. Consider any representable set A. By Problem 24, the set of all n such that $(n * 52) \in A$ is representable, say by Γ_1. By Problem 20, there is some term X (let n be its Gödel number) such that $\Gamma_1\lceil X\rceil = X$ holds. Consider the sentence $X = t$. We show that this is a Gödel sentence for A. On the one hand, suppose that $X = t$ is true. Then $\Gamma_1\lceil X\rceil = t$ also holds, meaning $n * 52 \in A$, that is, the Gödel number of $X = t$ is in A. On the other hand, suppose that $n * 52 \in A$. Then $X = \Gamma_1\lceil X\rceil = t$, that is, $X = t$ holds. Hence $X = t$ is a Gödel sentence for A.

Problem 26

Suppose that $\mathcal{T}_0$ was computable. By Problem 22 (2), it follows that $\overline{\mathcal{T}_0}$ is representable. By Problem 25, it would follow that there is some Gödel sentence X for $\overline{\mathcal{T}_0}$. That is, X is true iff the Gödel number of X is in $\overline{\mathcal{T}_0}$ iff X is not true. Thus X is true iff it is not true, which is a contradiction. Hence $\overline{\mathcal{T}_0}$ cannot be representable, and, consequently, $\mathcal{T}_0$ cannot be computable.